

2 Complex numbers	2.1	Solve any quadratic equation with real coefficients. Solve cubic or quartic equations with real coefficients.	Given sufficient information to deduce at least one root for cubics or at least one complex root or quadratic factor for quartics, for example: (i) $f(z) = 2z^3 + 5z^2 + 7z + 10$ Given that $z + 2$ is a factor of $f(z)$, use algebra to solve $f(z) = 0$ completely. (ii) $g(x) = x^4 - x^3 + 6x^2 + 14x - 20$ Given $g(1) = 0$ and $g(-2) = 0$, use algebra to solve $g(x) = 0$ completely.
	2.2	Add, subtract, multiply and divide complex numbers in the form $x + iy$ with x and y real. Understand and use the terms 'real part' and 'imaginary part'.	Students should know the meaning of the terms, 'modulus' and 'argument'.

Topic	What students need to learn:		
	Content	Guidance	
2 Complex numbers <i>continued</i>	2.3	Understand and use the complex conjugate. Know that non-real roots of polynomial equations with real coefficients occur in conjugate pairs.	Knowledge that if z_1 is a root of $f(z) = 0$ then z_1^* is also a root.
	2.4	Use and interpret Argand diagrams.	Students should be able to represent the sum or difference of two complex numbers on an Argand diagram.
	2.5	Convert between the Cartesian form and the modulus-argument form of a complex number.	Knowledge of radians is assumed.
	2.6	Multiply and divide complex numbers in modulus argument form.	Knowledge of the results $ z_1 z_2 = z_1 z_2 , \quad \left \frac{z_1}{z_2} \right = \frac{ z_1 }{ z_2 }$ $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$ Knowledge of radians and compound angle formulae is assumed.
	2.7	Construct and interpret simple loci in the argand diagram such as $z - a > r$ and $\arg(z - a) = \theta$.	To include loci such as $z - a = b$, $z - a = z - b$, $\arg(z - a) = \beta$, and regions such as $z - a \leq z - b$, $z - a \leq b$, $\alpha < \arg(z - a) < \beta$ Knowledge of radians is assumed.

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2 Complex numbers <i>continued</i>	2.8	Understand de Moivre's theorem and use it to find multiple angle formulae and sums of series.	To include using the results, $z + \frac{1}{z} = 2 \cos \theta$ and $z - \frac{1}{z} = 2i \sin \theta$ to find $\cos p\theta$, $\sin q\theta$ and $\tan r\theta$ in terms of powers of $\sin \theta$, $\cos \theta$ and $\tan \theta$ and powers of $\sin \theta$, $\cos \theta$ and $\tan \theta$ in terms of multiple angles. For sums of series, students should be able to show that, for example, $1 + z + z^2 + \dots + z^{n-1} = 1 + i \cot\left(\frac{\pi}{2n}\right)$ where $z = \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)$ and n is a positive integer.
	2.9	Know and use the definition $e^{i\theta} = \cos \theta + i \sin \theta$ and the form $z = re^{i\theta}$	Students should be familiar with $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$
	2.10	Find the n distinct n th roots of $re^{i\theta}$ for $r \neq 0$ and know that they form the vertices of a regular n -gon in the Argand diagram.	
	2.11	Use complex roots of unity to solve geometric problems.	